BABU BANARASI DAS UNIVERSITY

Assignment 1: IPL-Data-Analysis

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Instructions

Read the question carefully and try to understand what exactly is being asked.

About:

INDIAN PREMIER LEAGUE ®

The Indian Premier League (IPL) is one of the world's most popular and lucrative Twenty20 cricket leagues. Featuring 10 franchise-based teams from major Indian cities, it runs annually from March to May and has attracted over 500 million viewers per season since its inception in 2008. This tabular dataset includes match-by-match records from all IPL seasons, with details such as teams, player stats, scores, venues, toss decisions, win outcomes, and more.

Business Problem:

Analyze the given **Indian Premier League (IPL)** datasets and generate meaningful visual insights that could help IPL organizers and teams understand match dynamics, team and player performance, and audience-driven trends. Use these insights to identify what types of matches, teams, players, and venues generate higher engagement, and how the IPL can strengthen and grow its popularity across different cities, seasons, and regions.

Dataset:

match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bowler	is_super_over	...	bye_runs	legbye_runs	noball_runs	penalty_runs	batsman_runs	extra_runs	total_runs
1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	0	0	0
1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	0	0	0
1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	4	0	4
1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	0	0	0
1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	TS Mills	0	...	0	0	0	0	0	2	2

id	season	city	date	team1	team2	toss_winner	toss_decision	result	dl_applied	winner	win_by_runs	win_by_wickets	player_of_match	venue	umpire1	umpire2
1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	Sunrisers Hyderabad	35	0	Yuvraj Singh	Rajiv Gandhi International Stadium, Uppal	AY Dandekar	NJ Llong
2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	normal	0	Rising Pune Supergiant	0	7	SPD Smith	Maharashtra Cricket Association Stadium	A Nand Kishore	S Ravi
3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	normal	0	Kolkata Knight Riders	0	10	CA Lynn	Saurashtra Cricket Association Stadium	Nitin Menon	CK Nandan
4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	normal	0	Kings XI Punjab	0	6	GJ Maxwell	Holkar Cricket Stadium	AK Chaudhary	C Shamshuddin
5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	normal	0	Royal Challengers Bangalore	15	0	KM Jadhav	M Chinnaswamy Stadium	NaN	NaN

Link: Dataset_link(click here)

(***After clicking on the above link, you can download the files by right-clicking on the page and clicking on "Save As", then naming the file as per your wish, with .csv as the extension.***)

Dataset Details:

The datasets provided to you contain detailed information about **Indian Premier League (IPL)** matches and ball-by-ball events. These datasets can be used together to analyze match outcomes, team strategies, and player performance across different seasons.

Data Analysis with IPL match-by-match dataset from the seasons 2008 to 2019.
IPL dataset contains two files: **deliveries.csv** and **matches.csv**

1. Match Dataset

This dataset contains **match-level information** for each IPL game.

- **id**: Unique identifier for each match
- **season**: IPL season in which the match was played
- **city**: City where the match took place
- **date**: Date on which the match was played
- **team1**: First team participating in the match
- **team2**: Second team participating in the match
- **toss_winner**: Team that won the toss
- **toss_decision**: Decision taken after winning the toss (bat or field)
- **result**: Match result type (normal, tie, no result, etc.)
- **dl_applied**: Indicates whether the Duckworth–Lewis method was applied
- **winner**: Team that won the match
- **win_by_runs**: Margin of victory by runs (if applicable)
- **win_by_wickets**: Margin of victory by wickets (if applicable)
- **player_of_match**: Player awarded Player of the Match
- **venue**: Stadium where the match was played
- **umpire1**: First on-field umpire
- **umpire2**: Second on-field umpire
- **umpire3**: Third umpire (if available)

2. Ball-by-Ball (Deliveries) Dataset

This dataset contains **detailed ball-by-ball information** for each match.

- **match_id**: Unique identifier linking the delivery to a match
- **inning**: Inning number (1 or 2)
- **batting_team**: Team batting on that delivery
- **bowling_team**: Team bowling on that delivery
- **over**: Over number
- **ball**: Ball number within the over
- **batsman**: Batsman facing the delivery
- **non_striker**: Batsman at the non-striker's end
- **bowler**: Bowler delivering the ball
- **is_super_over**: Indicates whether the delivery was part of a super over
- **wide_runs**: Runs scored due to wides
- **bye_runs**: Runs scored as byes
- **legbye_runs**: Runs scored as leg byes
- **noball_runs**: Runs scored due to no-balls
- **penalty_runs**: Penalty runs awarded
- **batsman_runs**: Runs scored by the batsman
- **extra_runs**: Total extra runs on the delivery
- **total_runs**: Total runs scored on the delivery
- **player_dismissed**: Name of the dismissed player (if any)
- **dismissal_kind**: Type of dismissal (bowled, caught, run out, etc.)
- **fielder**: Fielder involved in the dismissal (if applicable)

Business-Oriented Analysis Questions

i) Match won by the maximum margin of runs.
ii) Match won by maximum wickets.
iii) Match won by the minimum margin of runs.
iv) Match won by minimum wickets.
v) Matches where D/L method was and wasn't applied.
vi) No. of matches held in each city.
vii) No. of matches won by each team.
viii) No. of matches held every season.
ix) Top 10 players based on no. of Man of Match (MOM) awards won.
x) Does winning the toss mean winning the match?
xi) What was the decision taken by captains when they won the toss?
xii) No. of matches where D/L method was applied every season.
xiii) Different results for matches.
xiv) How many times did a team win the toss?
xv) Best venue for defending and chasing a total.
xvi) Best defending and chasing teams.
xvii) No. of matches played in different stadiums.

Apart from this, feel free to do your own analysis and find new insights based on the data which is shared.

Submission Process

For Online (google Form):

1. Write the insights after writing the code and recommendations/insights in the same **IPYNB** file.
2. Convert your jupyter notebook into PDF (Save as PDF using Chrome browser's Print command) or simply click(ctrl + P) and it will show you the download option
3. adding images/graphs in the Ipynb file is compulsory by taking screenshots.
4. Upload both IPYNB file and PDF File In the Google Form (The link is mentioned below).

And

For Offline (Handwritten - Hard Copy):

1. Content Format

- Write the code neatly by hand on A4-sized sheets.
- Paste the output screenshot (printed and cut properly) right after each block of code.
- For each analysis, write insights/observations either:
    - As bullet points, or
    - In paragraph form (your choice).
- Make sure the insights are relevant and meaningful based on the output.

2. File Arrangement

- Use only A4-size sheets.
- Staple the sheets three times:
    - Once at the top left,
    - Once at the middle left, and
    - Once at the bottom left.
- Keep all sheets securely placed in a file/folder (plastic or paper file).

3. First Page Format

On the first page, write the all the necessary details clearly:

Like: University Name, Your Full Name, University Roll Number, Class Rollno, subject name etc.

Project Deadline: **09th March 2026 EOD**

Project Assignment Link:
https://docs.google.com/document/d/1540yQnSwkzYdcUam3xGao3Ge5qe0g7vxH5MNDhlA4nI/edit?usp=sharing

Project Submission Link:
https://forms.gle/oB6VH8X5tit8DfYc8

Thank You